

EXERCISE3: Stored Procedure

Creating Tables

```
-- =====
-- Create Accounts Table
-- =====

CREATE TABLE Accounts (
    AccountID NUMBER PRIMARY KEY,
    CustomerName VARCHAR2(50),
    AccountType VARCHAR2(20),
    Balance NUMBER(10,2)
);

-- =====
-- Create Employees Table
-- =====

CREATE TABLE Employees (
    EmployeeID NUMBER PRIMARY KEY,
    EmployeeName VARCHAR2(50),
    Department VARCHAR2(30),
    Salary NUMBER(10,2)
);

-- =====
-- Insert Sample Data
-- =====

INSERT INTO Accounts VALUES (101,'Nehal','Savings',10000);
INSERT INTO Accounts VALUES (102,'Rahul','Savings',15000);
INSERT INTO Accounts VALUES (103,'Priya','Current',20000);
INSERT INTO Accounts VALUES (104,'Amit','Savings',12000);

INSERT INTO Employees VALUES (1,'Rohit','IT',50000);
INSERT INTO Employees VALUES (2,'Sneha','HR',45000);
INSERT INTO Employees VALUES (3,'Karan','IT',60000);
INSERT INTO Employees VALUES (4,'Anjali','Finance',55000);
COMMIT;
```

Scenario 1:

```
-- =====
-- Scenario 1
-- Process Monthly Interest
-- =====

CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
AS
BEGIN
    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

    COMMIT;
END;
/

BEGIN
    ProcessMonthlyInterest;
END;
/

SELECT * FROM Accounts;
```

Scenario 2:

```
-- =====
-- Scenario 2
-- Update Employee Bonus
-- =====

CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (
    p_Department IN VARCHAR2,
    p_BonusPercent IN NUMBER
)
AS
BEGIN
```

```

    UPDATE Employees
    SET Salary = Salary + (Salary * p_BonusPercent / 100)
    WHERE Department = p_Department;

    COMMIT;
END;
/

BEGIN
    UpdateEmployeeBonus('IT',10);
END;
/

SELECT * FROM Employees;

```

Scenario 3:

```

-- =====
-- Scenario 3
-- Transfer Funds
-- =====

CREATE OR REPLACE PROCEDURE TransferFunds(
    p_FromAccount NUMBER,
    p_ToAccount NUMBER,
    p_Amount NUMBER
)
AS
    v_Balance NUMBER;
BEGIN
    SELECT Balance
    INTO v_Balance
    FROM Accounts
    WHERE AccountID = p_FromAccount;

    IF v_Balance >= p_Amount THEN

        UPDATE Accounts

```

```

        SET Balance = Balance - p_Amount
        WHERE AccountID = p_FromAccount;

        UPDATE Accounts
        SET Balance = Balance + p_Amount
        WHERE AccountID = p_ToAccount;

        COMMIT;

    END IF;
END;
/

BEGIN
    TransferFunds(101,102,2000);
END;
/

SELECT * FROM Accounts;

```

Output:

Query result Script output DBMS output Explain Plan SQL history					
  Download ▼ Execution time: 0.007 seconds					
	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE	
1	101	Nehal	Savings	10100	
2	102	Rahul	Savings	15150	
3	103	Priya	Current	20000	
4	104	Amit	Savings	12120	